

Information loss of fixed linear health indicators under competing degradation mechanisms and variable operating conditions

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Abstract

Predictive maintenance often estimates remaining useful life from a health indicator: a fixed-weight combination of condition-monitoring signals. When a component degrades through several mechanisms at different rates, the combination most informative about remaining life, the informative direction, changes with age, and no fixed indicator can follow it. This paper quantifies the information such an indicator loses and whether the operating point, which sets those rates, can reduce it. For units differing only in age, the loss depends on one angle between the indicator and the informative direction and fixes the increase in the error variance of the remaining-life estimate. If that direction turns through an arc Φ over life, a well-placed fixed indicator increases the variance by at most a factor $1/\cos^2(\Phi/2)$. On 129 lithium-ion cells the arc is 30.7° , so by at most 7.5%, and remaining-life estimates and replacement decisions are as good with a fixed indicator as with all signals. In simulation, adjusting the operating point reduces this already small loss 47-fold but barely improves the estimate, while lifetime varies 19-fold. Unless the arc is large, a fixed indicator is adequate and the operating point is better chosen for lifetime.

Keywords: health indicator, remaining useful life, prognostics, predictive maintenance, competing degradation mechanisms, operating conditions, information loss

1. Introduction

Predictive maintenance replaces or repairs a component shortly before it is expected to fail, rather than on a fixed schedule or after failure. The expectation comes from an estimate of the component's remaining useful life, and that estimate is in turn computed from condition monitoring. Monitoring seldom passes its raw measurements to the estimation algorithm: channels such as vibration, temperature, impedance and capacity are first combined into a single number, the health indicator, and the algorithm sees only that number [1, 2]. In practice the weights of the combination are usually fixed when the monitoring system is designed, because they are embedded in firmware, frozen by certification, or impractical to re-tune in the field. The indicator therefore determines how accurately the remaining useful life can be estimated and, through that estimate, how early maintenance must be scheduled to keep the risk of failure acceptable.

A fixed weighting is not always a good weighting. Engineering components commonly degrade through several mechanisms at once, and these mechanisms need not progress at the same rate. In a lithium-ion cell, for example, capacity fade and the growth of internal resistance

arise from different physical processes and need not evolve in proportion; one mechanism may dominate early in life while another accelerates towards failure. It helps to picture the monitoring channels as the axes of a space. As the component ages, its readings move through this space, and the direction in which they move is the combination of channels that changes most with age, and hence the one that says most about the remaining life. When the mechanisms progress at different rates, this direction turns as the component ages. A fixed indicator is a fixed direction in the same space, so it can be aligned with the most informative combination at one moment only, and the information it discards at other moments cannot be recovered by any prognostic algorithm that uses the indicator [3].

Two practical questions follow. The first is how much information a fixed indicator actually loses, and whether this can be assessed before the indicator is designed, when a different weighting or an adaptive indicator can still be chosen. The second concerns the operating conditions, such as temperature, load or duty cycle. These vary, either because the duty imposes them or because the operator selects them, and in both cases they change the relative rates of the mechanisms and hence the direction in which the readings move. Operators already select operating points to extend life [4, 5, 6]; the question is whether the operating point could also be chosen to keep a fixed indicator aligned, and whether doing so would be worthwhile.

Existing work addresses the first question from the per-

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spective of indicator construction, with criteria that differ from the one used here. Health indicators are constructed from data so that they change steadily and predictably with age, as measured by criteria such as monotonicity and trendability [2, 7], or by fusing sensors into a composite index whose weights are fitted once, for example so that the fused index has a consistent value at failure or follows a prescribed degradation model [8, 9, 10, 11, 12]; such indices have also been built under several operating conditions [13] and learned by neural networks [14], and reviews survey the resulting pipelines [1, 15]. Once fitted, these indices are fixed and linear in the channels, which is precisely the class of indicators considered in this paper, but none of the construction criteria measures how much information about remaining life the index discards as the mechanisms diverge. State-space fusion [16] avoids a fixed weighting but tracks a single degradation state, in which case the direction cannot turn. Section 4 compares one fusion criterion with the design proposed here on measured cells.

The second question connects to degradation modelling and to operation. Degradation under time-varying operating conditions is well modelled [17, 18, 19, 20, 21, 22, 23] and several degradation characteristics are known to respond differently to the same environment [24] or to depend on one another [25]. A separate body of work selects the operating point, or the load or production rate it implies, to extend life or to balance performance against reliability [4, 5, 6, 26, 27, 28], and prognostic accuracy has been assessed through its consequences for predictive maintenance [29]. These studies treat the monitoring signal as given; none asks what the operating point does to the information a fixed indicator retains. Our previous work bounded the information that any fixed compression of the channels retains along a single, given degradation trajectory [3], and the number of run-to-failure units required to learn such a compression [30]. The present paper builds on that analysis: it expresses the loss of a single linear indicator through one angle, relates the loss to the error of the remaining-life estimate and to maintenance decisions, bounds it by how far the informative direction turns, tests the bound on measured data, and lets the trajectory change with the operating point.

The main contributions of this paper are as follows.

- The information discarded by a fixed linear indicator is shown to be an exact function of a single misalignment angle, and the resulting inflation of the error variance of the remaining-life estimate, and of the safety margin of a predictive replacement policy, is obtained in closed form (Section 2).
- The arc traversed by the informative direction over the service life is shown to bound this inflation before any indicator is designed, which yields a five-step design procedure that requires run-to-failure data only (Section 3).

- In a case study on measured battery cells the bound is 7.5%, which corresponds to a safety margin of a predictive replacement policy at most 3.7% wider; a leave-one-out remaining-life estimator and a replacement policy using the fixed indicator perform as well as ones using all channels (Section 4).
- Selecting the operating point to keep the indicator aligned reduces the information loss substantially but leaves the error of a practical remaining-life estimator essentially unchanged, whereas the operating point alters the lifetime by an order of magnitude (Section 5). Public run-to-failure data support the premise that the informative direction turns but do not permit a test of the effect of the operating point on degradation (Section 6).

The remainder of the paper proceeds from the general to the specific. Section 2 states the degradation model, defines the information loss and lists the assumptions. Section 3 bounds the loss by how far the informative direction turns and derives the design procedure that follows from the bound. Section 4 applies the procedure to measured lithium-ion cells, up to a maintenance decision. Section 5 examines, on a synthetic system, whether selecting the operating point is worthwhile, and Section 6 examines the assumptions on public run-to-failure data. Section 7 answers the two questions and discusses the implications. Derivations are given in Appendix A.

2. Problem formulation

This section states how the mechanisms degrade and how they are observed (Sections 2.1 and 2.2), defines the information that a fixed indicator loses and relates it to the error of the remaining-life estimate (Section 2.3), and closes with the assumptions on which the rest of the paper rests. Figure 1 gives an overview of the construction.

2.1. Degradation model with operating-point dependence

A component degrades through d mechanisms, with damage states $x = (x_1, \dots, x_d)$, $x_i \geq 0$. A single operating point $u(t) \in \mathcal{U} \subset \mathbb{R}^m$, comprising m adjustable variables such as temperature, load and duty cycle, drives all of them; the envelope $\mathcal{U}$ is the bounded range of operating points allowed in service. The damage of each mechanism grows at a rate that depends on the operating point and on the damage already accumulated:

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d. \quad (1)$$

The stress factor $\varphi_i(u) > 0$ expresses how much faster mechanism i runs at u than at a reference operating point, and the shape function $h_i(x_i) > 0$ describes how its rate changes as damage accumulates. The two effects enter as separate factors, which makes the model separable. This is the standard accelerated-life form [17]: in the Arrhenius model $\varphi_i(u) = A_i \exp(-E_i/(kT))$, with activation

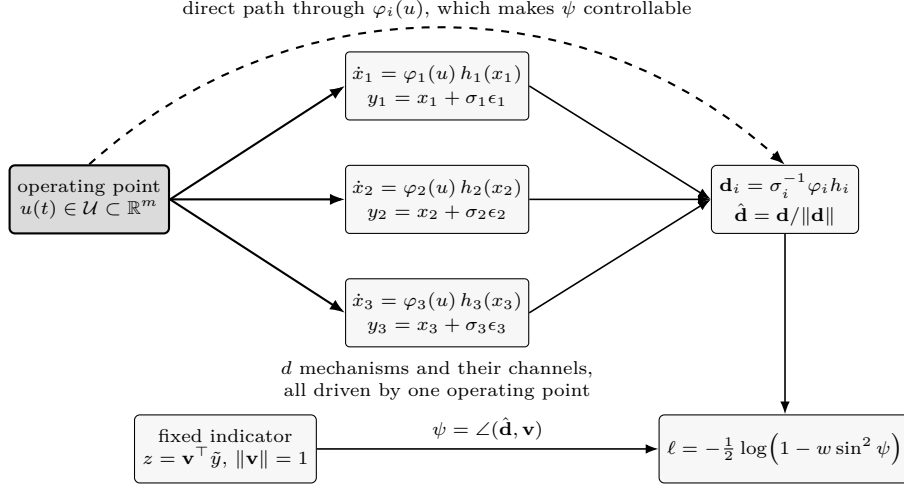


Figure 1: Overview of the problem formulation. One operating point drives all d mechanisms, each observed through its own channel. The informative direction $\hat{\mathbf{d}}$ is the weighting of the whitened channels that responds most strongly to the remaining life; the loss ℓ of a fixed indicator $\mathbf{v}$ depends only on the angle ψ between them. The operating point affects $\hat{\mathbf{d}}$ both through the damage and directly through the stress factors $\varphi_i(u)$; the direct path makes ψ controllable.

energy E_i , absolute temperature T and Boltzmann constant k , and suitable choices of h_i yield exponential, power law and stretched-exponential degradation [3]. Damage is normalised such that each mechanism exhausts the component at unity, and the component is withdrawn from service when the first mechanism reaches a condemnation limit $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (2)$$

The lifetime T_f depends on the operating point, and the normalised life $\tau = t/T_f$ runs from 0 at the start of service to 1 at withdrawal.

2.2. Remaining useful life and the informative direction

The quantity to be estimated is the remaining useful life $R(t) = T_f - t$. For a single unit with known parameters, (1) would determine it exactly; the uncertainty arises because the units of a fleet differ [17]. The analysis assumes that they differ only in age: all units follow a common degradation path, and each is at a different position along it, as are identical components installed at different times. Such a population is called a time-shift population, and R is taken to be Gaussian over it with variance $\text{Var}[R]$. For two units of such a population whose remaining lives differ slightly, the damage states differ, to first order, only along the direction in which the damage is currently growing. Since R decreases at unit rate, the sensitivity of mechanism i to the remaining life is therefore

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (3)$$

Each mechanism is observed through a dedicated channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (4)$$

with independent standard normal noise ϵ_i and noise level σ_i . This pairing of channels with mechanisms is an assumption: condition monitoring in general yields $y = H(x, u) + \epsilon$, with sensors that mix mechanisms, and Section 6 shows that raw turbofan sensors violate it. To make channels with different units and noise levels comparable, each is divided by its noise level, which is called whitening; the whitened readings $\tilde{y} = (y_i/\sigma_i)$ at one inspection are called the record. By (3), the record responds to the remaining life through the signal vector

$$\mathbf{d} = [\sigma_i^{-1} \varphi_i(u) h_i(x_i)]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d} / \|\mathbf{d}\|. \quad (5)$$

The norm $\|\mathbf{d}\|$ measures how strongly the record responds to the remaining life, and the unit vector $\hat{\mathbf{d}}$, a point on the unit sphere of the d -dimensional space of whitened channels, is the combination of channels along which it responds. It is referred to as the *informative direction*: the weighting of the channels that, at the current state, carries the most information about the remaining life. All entries of $\mathbf{d}$ are positive, and $\hat{\mathbf{d}}$ depends on both the state and the operating point. The operating point affects $\hat{\mathbf{d}}$ through two paths: indirectly, through the damage it produces, and directly, through $\varphi_i(u)$.

2.3. Information loss of a fixed linear indicator

A fixed linear indicator combines the record with a weight vector selected once at the design stage:

$$z = \mathbf{v}^\top \tilde{\mathbf{y}}, \quad \|\mathbf{v}\| = 1. \quad (6)$$

The weight vector is a second point on the same unit sphere. An indicator defined on raw measurements, with weights $\mathbf{a}$, is the same mapping with $\mathbf{v} \propto \Sigma^{1/2} \mathbf{a}$, where $\Sigma = \text{diag}(\sigma_i^2)$. Let ψ denote the angle between $\mathbf{v}$ and $\hat{\mathbf{d}}$, called the misalignment, and let $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ denote

the signal-to-noise ratio of the record, that is, the spread of the remaining life across the population as seen through the record, relative to the noise. Linearising the record about the nominal path renders it jointly Gaussian with R ; the indicator then retains the fraction $\cos^2 \psi$ of the signal and all of the noise.

Information is measured here by the mutual information between the remaining life and the observation, in natural units (nats), and the loss ℓ is the information about R carried by the record but not by the indicator. It is therefore (Appendix A.1)

$$\ell = \frac{1}{2} \log \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (7)$$

with $w = \gamma/(1 + \gamma)$. The loss has a direct statistical meaning: the error variance of the optimal estimate of R based on the indicator exceeds that based on the record by the factor

$$\frac{\text{Var}[R | z]}{\text{Var}[R | \tilde{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}. \quad (8)$$

A loss of 0.01 nat, for instance, means an error variance about 2% larger than with all channels, and this excess, $e^{2\ell} - 1$, is the quantity that the rest of the paper reports alongside the loss. The loss depends on the indicator only through the single angle ψ , and for small misalignment $\ell \approx \frac{w}{2} \psi^2$. The design objective is its average over the service life,

$$\bar{\ell} = \frac{1}{T_f} \int_0^{T_f} \ell(t) dt. \quad (9)$$

The same reduction applies when the indicator is read at many inspections, with γ summed over the records of the inspections and $\sin^2 \psi$ averaged with weights $\|\mathbf{d}\|^2$ (Supplementary Section S2).

The loss also bears directly on maintenance decisions. Under the Gaussian model, a predictive replacement policy that replaces a unit once its probability of failing within a lead time t_{lead} reaches α does so when the estimated remaining life falls to $t_{\text{lead}} + \zeta_\alpha s_R$, where s_R is the standard deviation of the remaining life given the observation and ζ_α the standard normal quantile of order $1 - \alpha$. The safety margin $\zeta_\alpha s_R$ is, on average, remaining life discarded at replacement beyond the lead time, and a fixed indicator multiplies it by e^ℓ relative to the record. Decision-oriented assessments of prognostic algorithms weigh precisely this margin against the risk of failure [29].

A further consequence concerns the design of tests and sensors. Adding or improving sensors increases the information in the record, but does not by itself reduce the loss of a fixed indicator. The design of accelerated degradation tests and of inputs for estimability usually maximises the Fisher information of the record [31], here $I_{\text{rec}} = \|\mathbf{d}\|^2$ about R when the prior variance is normalised to one; the loss, however, is determined by this quantity together with the Fisher information of the indicator itself, $I_z = (\mathbf{v}^\top \mathbf{d})^2$.

$$\ell = \frac{1}{2} \log(1 + \text{Var}[R] I_{\text{rec}}) - \frac{1}{2} \log(1 + \text{Var}[R] I_z). \quad (10)$$

For a fixed value of I_{rec} the loss still ranges from zero to $\frac{1}{2} \log(1 + \gamma)$ as ψ varies. Maximising the information in the record therefore does not minimise the loss of a fixed indicator, whereas including I_z in the criterion does. Supplementary Section S4 shows this on the synthetic system of Section 5.2.

2.4. Assumptions

The analysis rests on four assumptions, to which the later sections refer.

- (A1) Time shift: the units of a population follow a common degradation path and differ only in their position along it.
- (A2) Linear-Gaussian record: the remaining life is Gaussian over the population, and the record is linearised about the nominal path.
- (A3) Channel pairing: each mechanism is observed through its own channel with independent Gaussian noise, as in (4).
- (A4) Separable stress: the operating point and the accumulated damage affect each mechanism's rate as separate factors, as in (1).

The loss (7) and the bound of Section 3 need (A1) and (A2), and hold for the direction in which the observed channels respond, whether or not each channel tracks a single mechanism. (A3) and (A4) are needed only for the analysis of the operating point in Section 5. Section 4 states which assumptions the measured cells satisfy, and Section 6 tests them on public data.

3. Rotation of the informative direction and a bound on the loss

Section 2 showed that the loss of a fixed indicator depends only on its angle to the informative direction. How large that angle must become over a life depends on how far the informative direction turns. This section explains why it turns, bounds the loss by the length of its path, and derives from the bound a design procedure.

3.1. Rotation rate of the informative direction

As the component ages, the tip of the unit vector $\hat{\mathbf{d}}$ traces a path on the unit sphere. Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i and $p_i = \hat{\mathbf{d}}_i^2$ the share of the record's Fisher information carried by channel i . Differentiating (5) along (1) gives (Appendix A.2)

$$\|\dot{\hat{\mathbf{d}}}\| = \text{Std}_p[\dot{g}], \quad \dot{g}_i = \varphi_i(u) h'_i(x_i) + \nabla_u \log \varphi_i(u) \cdot \dot{u}, \quad (11)$$

where Std_p is the standard deviation over channels weighted by the shares p_i . The informative direction thus turns at a speed equal to the information-weighted spread of the logarithmic rates of the mechanisms. If all mechanisms accelerate in proportion, the direction stays fixed however rapidly the component degrades; it turns

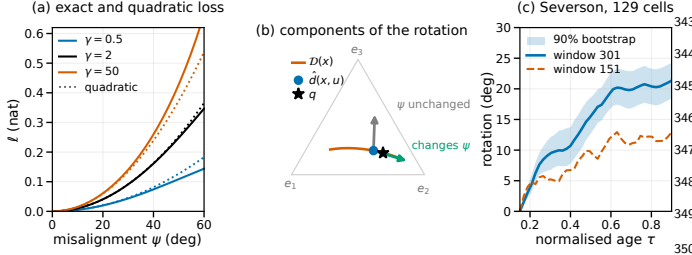


Figure 2: (a) The exact loss (7) (solid) and its quadratic approximation (dotted); the approximation overstates the loss when $\gamma < 2$ and understates it when $\gamma > 2$. (b) Only the component of the rotation of the informative direction along the great circle towards the indicator changes the misalignment. (c) Rotation of the observed direction on the Severson battery cells (Section 4), estimated with smoothing windows of 301 and 151 cycles.

only because the mechanisms disagree. The second term of $\hat{g}_i$ represents the influence of the operating point, which Section 5 exploits: a change in the operating point alters the rates that turn the direction. Not every turn matters for a given indicator. Only the component of the motion along the great circle towards $\mathbf{v}$, the shortest path on the sphere between $\hat{\mathbf{d}}$ and $\mathbf{v}$, changes ψ ; the remaining $d - 2$ components move the direction without affecting the information retained by the indicator (Fig. 2(b)).

3.2. A bound in terms of the rotation arc

Suppose that over a window of the service life the informative direction traverses a path of total length $\Phi < \pi$ on the sphere, called its arc. The direction at the midpoint of this arc lies within $\Phi/2$ of every point on it, so a fixed indicator placed at the midpoint satisfies $\psi \leq \Phi/2$ throughout. Since the loss increases with ψ , such an indicator satisfies at every instant,

$$e^{2\ell} \leq \frac{1}{1 - w \sin^2(\Phi/2)} \leq \frac{1}{\cos^2(\Phi/2)}. \quad (12)$$

The same holds for any fixed indicator whose largest angle to the path, $\psi_{\max}$, is at most $\Phi/2$. The bound requires only the arc, which can be estimated from run-to-failure data before any indicator is designed. An arc of 30° allows at most 7.2% additional error variance, 60° at most 33%, and 90° a doubling; the right-hand bound corresponds to the limit of high signal-to-noise ratio and is independent of γ . Correspondingly, the safety margin of the replacement policy of Section 2.3 grows by at most the factor $1/\cos(\Phi/2)$. The best fixed indicator lies close to the midpoint of the arc; Supplementary Section S2 characterises it as a barycentre of the path of the informative direction. Figure 2(a) shows how the loss grows with the misalignment angle, and Fig. 2(c) the rotation measured on the battery cells of Section 4.

3.3. Design procedure

The bound yields a design procedure that requires run-to-failure data only.

1. Estimate the informative direction from the fleet: at each value of the normalised life τ , take the time derivative of the monitoring channels, whiten each channel by its measurement noise, estimated from the scatter of the channel about a smoothed trend, and average the resulting unit vectors over the units. The result is called the observed direction.
2. Compute the arc Φ that the observed direction traverses and the bound (12). The bound is the largest increase in error variance that a well-placed fixed indicator can cause; if it is small compared with the accuracy the maintenance decision requires, a fixed indicator suffices.
3. Place the fixed indicator at the barycentre of the path, in practice the renormalised mean of the observed direction, and check that its largest angle to the path does not exceed $\Phi/2$.
4. Validate the indicator with a leave-one-out remaining-life estimator, against the same estimator run on all channels.
5. Evaluate the maintenance decision that the estimate supports.

Section 4 carries out the five steps on measured lithium-ion cells.

4. Case study: lithium-ion cells

The procedure is applied to the lithium-ion cells of Severson et al. [32], a public fleet cycled to failure under different fast-charging protocols and widely used as a prognostics benchmark. The monitoring channels are discharge capacity, internal resistance and mean cell temperature. The 129 cells that meet the selection criteria of the protocol in Supplementary Section S1 have lives from 447 to 1932 cycles; because the lives differ more than fourfold across protocols, the population is far from a time shift. The cells therefore violate (A1), and the procedure treats them as if they differed only in age along a common path in normalised life; the leave-one-out test of step 4 checks whether that approximation is good enough in practice. (A3) and (A4) are not needed here. Supplementary Section S1 gives the protocol and parameters of every step.

4.1. Steps 1 to 3: rotation, bound and indicator

The observed direction of step 1, which estimates the informative direction as a fleet mean, moves through an arc of 30.7° between 15% and 90% of life (Fig. 2(c)). After whitening, the fixed indicator at its barycentre weights capacity, resistance and temperature as (0.99, 0.10, 0.12) and lies at most 14° from the informative direction, within half the arc. By (12), it therefore inflates the error variance of the remaining-life estimate by at most 7.5% over that window, irrespective of the signal-to-noise ratio, and widens the safety margin of a predictive replacement policy by at most 3.7%.

4.2. Step 4: remaining-life estimation

The bound concerns the optimal estimate given the record at one inspection. A practical estimator uses a history of inspections and is not optimal, so its error with the fixed indicator can fall on either side of its error with all channels; what the comparison tests is whether the information loss matters for such an estimator. A similarity-based estimator matches the last 30% of a cell's elapsed cycles against the fleet reference path in normalised life and returns the life that fits best; everything it uses, including the fixed indicator, is computed without the cell being tested. With all three channels its root-mean-square error is 148, 83 and 49 cycles at 50%, 70% and 90% of life. Run on the fixed indicator at the barycentre of the observed direction, the same estimator attains between 0.96 and 1.00 times the mean squared error of the three-channel estimator at these ages (Table 1). Capacity alone, the conventional battery health indicator, lies up to 22° from the informative direction, for which the bound allows 17° , and raises the mean squared error by 5% at half of life and by less later. Temperature alone, a deliberately misaligned control, raises it at least 40-fold, so the test is able to detect a poorly chosen indicator. A data-level fusion indicator, whose weights minimise the between-cell spread of the indicator at failure relative to its mean change over life, in the spirit of [8], lies 75° to 82° from the informative direction and has 28.9 times the mean squared error at half of life. Near failure, the stage its construction targets, it is the most accurate signal, at 0.81 times the three-channel error: it suppresses between-cell differences in the level of the channels at failure, a source of error that a time-shift population does not contain.

4.3. Step 5: replacement decisions

The last step evaluates the decision directly. Each cell is inspected every 25 cycles from cycle 100 and is replaced at the first inspection at which the predicted remaining life is at most a threshold of n_{thr} cycles; a cell not replaced before its end of life counts as a failure. For a failure ten times as costly as a preventive replacement, the threshold that minimises the long-run cost per cycle [29] gives the costs of Table 1. The fixed indicator, capacity alone and all three channels reach the same cost to within 0.1%, with 1.6% of cells failing and about 134 cycles, a fifth of a cell's life, left unused at replacement; the fusion indicator costs 2.5% more. For cost ratios from 5 to 50 the fixed indicator stays within 1% of the three-channel cost. The thresholds are chosen on the same cells for every input, so the absolute costs are optimistic, but the comparison between inputs is not. The case study thus confirms, on measured data and at the level of a maintenance decision, what the bound predicts: for this fleet a fixed indicator is adequate.

5. Selecting the operating point

The case study took the operating conditions as given. In many applications they are not: operators choose tem-

peratures, loads or charging rates, often to extend life. Because the operating point sets the relative rates of the mechanisms, it also moves the informative direction, and it could in principle be chosen to keep a fixed indicator aligned. This practice, referred to here as steering, is examined in this section. No public fleet lets the operating point act on the damage (Section 6), so the examination uses a synthetic system. Section 3 already limits what steering can achieve: the excess error variance that it can remove is at most the bound (12) for the arc of the unsteered trajectory.

5.1. Limits of steering

At a state x , the directions attainable through the operating point form the set

$$\mathcal{D}(x) = \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\}, \quad (13)$$

and the smallest misalignment achievable at x , called the floor, is $\beta_{\min}(x) = \angle(\mathbf{v}, \mathcal{D}(x))$. Two structural results, derived in Supplementary Section S4, delimit what steering can do. First, holding the misalignment at a constant, non-zero level is a single scalar condition and needs only one adjustable operating variable. Second, removing the loss altogether is far more demanding. If the informative direction equals $\mathbf{v}$ at every instant, the damage can only grow along the fixed direction $\Sigma^{1/2}\mathbf{v}$, so the state must move along a straight line in damage space. Keeping it on that line requires, except for special combinations of parameters, one fewer independent operating variable than there are mechanisms, together with an aligning input that stays within the allowed envelope $\mathcal{U}$ along the whole line. A component with three mechanisms and a single temperature setpoint can therefore reduce the loss but, generically, not remove it.

When the loss cannot be removed, the relevant quantity is the smallest achievable average loss,

$$\bar{\ell}^* = \min_{u(\cdot)} \frac{1}{T_f} \int_0^{T_f} \ell \, dt, \quad (14)$$

over all operating policies $u(\cdot)$, that is, all ways of setting the operating point over time. Both the integral and the lifetime T_f depend on the policy, so the objective is a ratio of two quantities that the operating point changes together, and the operating point that aligns the indicator is in general not the one that prolongs life. The simplest policy disregards the lifetime and applies, at each state, the input that minimises the misalignment; it is referred to as the myopic policy.

5.2. A numerical study

The synthetic system is chosen to resemble the thermally activated degradation of a battery with three mechanisms and one adjustable temperature. It has $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in which the input u is a scaled reciprocal temperature,

Table 1: Case study on the Severson cells. $\psi_{\max}$ is the largest angle between the estimator input and the fleet-mean informative direction over $\tau \in [0.15, 0.90]$; the mean-squared-error (MSE) ratios compare the leave-one-out remaining-life estimator run on the input with the same estimator run on all three channels, at 50%, 70% and 90% of life; the replacement columns give the minimal long-run cost per 1000 cycles, in units of a preventive replacement, when a failure costs ten times as much, with the fraction of cells failing and the mean life left unused at replacement, in cycles.

estimator input	$\psi_{\max}$	MSE ratio			replacement policy		
		50%	70%	90%	cost	failures	unused
all channels	—	1	1	1	1.689	1.6%	134
fixed indicator	14°	0.960	0.972	0.999	1.688	1.6%	133
capacity	22°	1.050	1.014	0.986	1.690	1.6%	134
fusion	82°	28.9	8.43	0.807	1.731	1.6%	150
temperature	90°	43.7	171	725	—	—	—

$u = 3(1/(kT) - 1/(kT_{60}))/((1/(kT_{25}) - 1/(kT_{60}))$ with T_{60} and T_{25} the absolute temperatures of 60 and 25 °C, so that $u = 0$ corresponds to 60 °C and $u = 3$ to 25 °C. The parameters are $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, which in these units are activation energies of 0.44, 0.81 and 0.59 eV, with shape functions $h_i(x) = 1 + c_i x$ and $c = (-0.5, 2.0, 0.8)$, which give one decelerating and two accelerating mechanisms. Time is measured in units of the characteristic time of the first mechanism at 60 °C where $\varphi_1 = 1$. There is a single input $u \in [0, 3]$, so by Section 5.1 the loss can be reduced but, generically, not removed; Supplementary Section S4 confirms this for the present system. The noise has unit variance and the prior variance is $\text{Var}[R] = 10^3$; the weight $w = \gamma/(1 + \gamma)$ of (7) is evaluated at the system's own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than held fixed. Every trajectory starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$; the values of the reported constants at $x_f = 1$ are given in Supplementary Section S4. The identities of Sections 2 and 3 were verified numerically on this system (Supplementary Section S3).

The bound of Section 3 can be checked first. Under constant inputs the arc is between 24.8° and 35.6°, the bound allows between 4.8% and 10.3%, and the best fixed indicator for each input incurs between 1.6% and 3.4%; the bound therefore holds and overestimates the excess by a factor of about three. These small excesses are all that steering can remove.

The comparison of strategies needs three indicators. The steering indicator q^* is the one for which the myopic policy achieves the smallest average loss; the balanced indicator weights the three whitened channels equally; and for each constant operating point, the constant design uses the indicator that is best for that point. With the input of the myopic policy searched on a grid of 41 points of $\mathcal{U}$, the steering indicator has the weight vector $\mathbf{v}^* = (0.466, 0.771, 0.435)$, at which the myopic policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input, a factor of 68. This comparison holds the indicator fixed at the one designed for steering. For operation without steering, the indicator would instead be designed for the constant input, and minimising over the indicator and the constant input jointly gives 7.95×10^{-3} , at the high-temperature end $u = 0$ of the envelope. Relative to

this baseline, the reduction achieved by steering is a factor of 47, the figure quoted in the abstract. The myopic policy is also close to the best achievable by any policy: the best policy for the ratio (14), computed on state grids of up to 137 points per axis, lies within 2.1% of it on every control grid, and both factors are stable under refinement of the grids (Supplementary Section S4).

Table 2 reports the practical consequences. By (8), a loss of 7.95×10^{-3} already corresponds to an error variance only 1.6% above that of the complete record, and steering removes almost all of this excess. A practical estimator, however, is largely insensitive to it. A time-shifted population is simulated along the path of each policy, each unit is observed at 30 instants over the last 40% of a lifetime before its remaining life is estimated, with unit channel noise, and the estimate is the time offset that best matches the reference path of the policy, computed once from the indicator and once from all three channels. Under every policy the two estimators differ by less than 1.2%, and the error as a fraction of life stays between 8.4% and 10.7%, being determined by the path rather than by the indicator. The operating point does, however, change the lifetime substantially. Across the constant designs of the table it varies 19-fold, because the mechanisms run faster when hot. Among the policies, the myopic policy at q^* attains a lifetime 1.66 times that of the best constant input at the same indicator and 3.8 times that of the constant design with the smallest loss; and a myopic policy aimed at the balanced indicator, which requires far less steering, extends the lifetime by a further factor of 3.4. For this system the choice of operating point matters far more for lifetime than for monitoring.

5.3. Sensitivity to model misspecification

Where run-to-failure data are scarce, the indicator can instead be designed from a degradation model, by applying steps 1 to 3 of the procedure to trajectories simulated from the model. Such an indicator inherits the errors of the model, and this subsection asks how accurate the model must be. For each error level ω , a perturbed energy vector $E(1 + \omega\eta)$ is drawn, with η standard normal, and the barycentre indicator is computed from trajectories of this misspecified system under a constant input. The indicator

Table 2: Information loss, remaining-life estimation error and lifetime of the synthetic system under different operating strategies. $\bar{\ell}$ is the information loss (9); $e^{2\bar{\ell}} - 1$ the corresponding excess error variance (8); the remaining-useful-life (RUL) errors are the median absolute error of a template-matching remaining-life estimator over 2000 units, as a percentage of life, based on the indicator or on all three channels; T_f is the lifetime. Constant designs use the indicator that is best for their own input.

operating strategy	$\bar{\ell}$	$e^{2\bar{\ell}} - 1$	RUL error (% of life)		T_f
			indicator	all channels	
constant, 60 °C	7.95×10^{-3}	1.6%	10.7%	10.8%	0.378
constant, 41.5 °C	1.44×10^{-2}	2.9%	8.9%	8.8%	1.967
constant, 25 °C	1.26×10^{-2}	2.6%	8.8%	8.7%	7.203
myopic, q^*	1.68×10^{-4}	0.03%	9.5%	9.5%	1.433
myopic, balanced	4.08×10^{-3}	0.8%	8.4%	8.5%	4.928

Table 3: Effect of designing the indicator from a misspecified stress model. Entries are the loss ratio: the loss on the true system relative to that of the indicator designed from the true model, over 40 draws per level; ω is the relative standard deviation of the error in the activation energies. The steering policy uses the true system.

ω	median	90th percentile	worst of 40
0.02	1.03	1.52	2.04
0.05	1.40	2.94	6.44
0.10	3.10	10.8	67.7
0.20	10.0	44.9	78.6
0.35	36.0	94.1	195

is then evaluated on the true system under the myopic policy aimed at it, with every loss evaluated at the system’s own γ . Since the policy is computed from the true system, the table isolates the effect of an error in the indicator; an error that also affected the policy would be more costly. Table 3 reports the loss ratio, the loss of this indicator divided by that of the indicator designed from the true model, over 40 draws per level; a ratio of 1 indicates that the error has no effect.

An error of 2% in the energies has a negligible effect. At 5% the median loss ratio is 1.40. At 10% the median loss ratio is 3.10 and the worst of forty draws reaches 67.7, comparable in magnitude to the entire benefit of steering; beyond this level every column increases monotonically, to a median of 36.0 at $\omega = 0.35$. The loss to which these ratios apply is small, however, so that even a tenfold increase leaves the indicator’s excess error variance below 1%.

6. Evidence from public run-to-failure data

The preceding sections rest on premises that public run-to-failure data can partly test. Three are examined here: that the informative direction rotates; that the operating point moves it by acting on the degradation, the actuation premise on which Section 5 relies; and that the separable stress model (A4), interpreted through the pairing of channels with mechanisms (A3), holds. The tests use the observed direction of step 1 of the procedure, estimated from data, which coincides with the informative direction only under (A1) and (A3). Table 4 summarises what public run-to-failure fleets establish and what they leave

open. The fleets are the Severson battery cells, the XJTU-SY and PRONOSTIA bearing test rigs, the NASA battery cells, and the C-MAPSS and N-CMAPSS turbofan engine simulations of NASA. Supplementary Section S1 gives the protocol behind every row, and its Table S1 lists the parameters of the code that produced it.

Rotation of the informative direction is observed on the battery cells of Section 4. That analysis and the turbofan contrasts below interpret a time derivative as the sensitivity (3), and therefore presuppose the time-shift population (A1).

The effect of the operating point on degradation remains untested. The measured fleets either do not vary the operating point within a unit or contain too few units. The two turbofan simulations do vary it, between discrete operating regimes of altitude, speed and throttle, but their generators preclude the test. C-MAPSS imposes its flow and efficiency losses as exponential functions of time, from a random initial deterioration, and lets the regime act only on the thermodynamic response of the sensors [36]. N-CMAPSS imposes a linear normal degradation followed by a steeper abnormal one, with the operating history entering through the onset of the abnormal phase [38], and the health parameters supplied with the data are constant within every flight. On C-MAPSS the observed directions of different regimes of the same engine differ by more (B) than those of two halves of one regime (W). This contrast, positive in all but three engines against a negative control that returns none, therefore shows that the observed direction depends on the regime through the sensor response alone. Variable operating conditions thus rotate the observed direction even when they do not act on the damage. Because the loss of a fixed indicator depends on the observed direction regardless of the cause of its motion, this finding is relevant to indicator design; it is not, however, evidence that the operating point acts on the damage.

The separable model fails on raw turbofan sensors. On both N-CMAPSS datasets the interaction between operating point and life stage exceeds the main effect of the operating point. The test rejects (A3) and (A4) jointly, and since the simulated damage does not respond to the operating point within a flight, the violation is attributable to the sensor map: the sensors mix mechanisms with weights that depend on the operating point, in violation of (4).

Table 4: Premises of the model examined on six public run-to-failure fleets. Only the two turbofan simulations vary the operating point within a unit, and neither allows it to act on the damage; the actuation premise is therefore untested on every public fleet. B and W are the mean angles between observed directions of different operating regimes and of two halves of the same regime.

fleet	premise	quantity measured	finding
Severson lithium-ion [32]	rotation	arc of an empirically constructed informative direction, the fleet mean, over $\tau \in [0.15, 0.9]$, on 129 Severson lithium-ion cells	travels 30.7° of arc and ends 21° from its start (Fig. 2(c)); all three weights stay positive once the smoothing window reaches 301 cycles
XJTU-SY [33], PRONOSTIA [34]	actuation	—	the operating condition is fixed per unit, so condition and group of units cannot be separated
NASA batteries [35]	actuation	—	only four cells
C-MAPSS [36]	actuation: not testable	between-regime minus within-regime angle, $B - W$, each engine serving as its own control	by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines; surrogate regimes give $ B - W \leq 0.5^\circ$; the damage is independent of the regime by construction, so the contrast reflects the sensor map
N-CMAPSS DS02, DS01 [37]	separability with pairing (4)	interaction against both main effects, in a two-way layout inside engines	larger by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, under every clustering and staging tried: rejected jointly

The pairing is what makes the informative direction positive, so the geometry of Sections 3 and 5 applies to such sensors only after the observation map has been inverted, and the offline screen of how far the operating point can move the informative direction (its actuation authority; Supplementary Section S4) is available only where the separable form has been shown to hold. The two-way layout, whose protocol is given in Supplementary Section S1, provides this test at the cost of a single pass over data that a reliability programme typically already holds.

7. Discussion and conclusions

The paper asked how much information about remaining life a fixed health indicator discards when degradation mechanisms progress at different rates, and whether the operating point can reduce the loss. The loss is set by how far the informative direction turns over the life, and on the battery cells examined here the turn is small enough that a fixed indicator supports remaining-life estimates and replacement decisions as well as all channels do. The operating point can reduce the loss, but when the turn is small there is little to gain.

The results have three practical implications.

First, the rotation arc should be assessed before an adaptive indicator is considered. The bound (12) converts a measured arc into a worst-case increase in the error variance of the remaining-life estimate. For the battery fleet examined here the resulting increase is at most 7.5%, and for the synthetic system the actual excess is about one-third of the bound. Where the arc amounts to a few tens of degrees, a single fixed indicator is adequate; indicators whose weights change with age, whether adaptive, segmented by life stage or learned, become worthwhile as the arc approaches a right angle, at which the bound allows the error variance to double. The five steps of Section 3.3 make this assessment from run-to-failure data alone.

Second, the operating point should be selected primarily for service life. Steering the operating point can reduce the information loss by an order of magnitude or more. Where the loss is already small, however, this yields almost no improvement in the accuracy of a practical remaining-life estimator, whereas the same choice alters the lifetime by an order of magnitude. Aligning an indicator and prolonging the life of the component are distinct objectives, and when the arc is small the latter is the more consequential.

Third, variable operating conditions should be represented when the arc is estimated. On the turbofan simulations the observed direction differs between operating regimes even though the damage does not respond to them, because the sensors respond differently in different regimes. An arc estimated under a single operating condition can therefore understate the rotation that a fixed indicator will meet in service.

Limitations. Four limitations qualify these conclusions. First, the construction assumes a time-shift population (A1); dispersing the degradation coefficients of the synthetic system rotates the population regression of damage on remaining life by about 45° away from the direction the construction uses, more than the measured arc of the battery cells, so the bound does not cover such dispersion, although the battery case study, whose cells violate (A1), suggests that the practical conclusion survives it. Second, the pairing of channels with mechanisms (A3) fails on raw turbofan sensors, so the geometry applies to them only after the observation map has been inverted. Third, the actuation premise remains untested on every public fleet. Finally, the best achievable loss is computed and bracketed between the myopic policy and an extrapolation of the computation to infinitely fine grids, not bounded analytically.

Appendix A. Derivations

Appendix A.1. Information loss and error-variance inflation

Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$, the linearised whitened record. The indicator returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$ whose noise is standard normal because $\|\mathbf{v}\| = 1$, so its signal-to-noise ratio is $\gamma \cos^2 \psi$. The record's sufficient statistic $\mathbf{d}^\top \tilde{y}$ has ratio γ . A scalar Gaussian channel of signal-to-noise ratio s carries $\frac{1}{2} \log(1 + s)$ nats about R [39], so the loss, the mutual information of R with the record minus that with the indicator, is the first form of (7); substituting $\cos^2 \psi = 1 - \sin^2 \psi$ gives the second. The posterior variances are $\text{Var}[R]/(1 + \gamma \cos^2 \psi)$ and $\text{Var}[R]/(1 + \gamma)$, whose quotient is (8). Equation (10) is (7) with $\gamma = \text{Var}[R]I_{\text{rec}}$ and $\gamma \cos^2 \psi = \text{Var}[R]I_z$; at fixed I_{rec} , w is fixed and ℓ increases continuously from zero towards $\frac{1}{2} \log(1 + \gamma)$ as ψ goes from 0 to $\pi/2$.

Appendix A.2. Rotation of the informative direction

From $\mathbf{d}_i = e^{g_i}$, $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, and $\dot{\mathbf{d}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has components $\dot{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$ with $p_i = \hat{\mathbf{d}}_i^2$, so its squared norm is $\text{Var}_p[\dot{g}]$. Differentiating $g_i = \log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (1) gives $\dot{g}_i$ in (11). With $\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v}$, $-\sin \psi \dot{\psi} = \dot{\mathbf{d}}^\top \mathbf{v}$, so only the component of $\dot{\mathbf{d}}$ along the great circle towards $\mathbf{v}$ changes ψ .

Data availability and reproducibility

The six fleets named in Section 6 are publicly available. One script recomputes every reported number from a single configuration and compares it with the value stated in this manuscript and in its supplementary material, covering the tables and the quantities reported alongside them; all of its checks pass, and the output of the run that verified this version is included. The random-system sweep and the downstream repetition of Supplementary Section S4 and the grid sequence of Section 5.2 require several hours of computation, so they are recorded by three further scripts; the checks read these records and recompute part of the grid sequence afresh. One group of numbers lies outside the checks: the residuals of the identity checks in Supplementary Section S3. The figures are produced by three further scripts. All of these are available at <https://github.com/HuyHoangLe0201/health-indicator-information-loss>, and are also provided with this article as a supplementary archive, together with the Supplementary Material.

CRedit authorship contribution statement

Huy Hoang Le: Conceptualization, Methodology, Formal analysis, Software, Validation, Investigation, Data curation, Visualization, Writing – original draft. **Kim Anh Nguyen:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision, Project administration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with drafting and editing the text and with writing and checking the analysis and verification code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, J. Lin, Machinery health prognostics: A systematic review from data acquisition to RUL prediction, *Mech. Syst. Signal Process.* 104 (2018) 799–834. doi:10.1016/j.ymssp.2017.11.016.
- [2] J. Coble, J. W. Hines, Identifying optimal prognostic parameters from data: A genetic algorithms approach, *Annual Conference of the PHM Society*, vol. 1, no. 1, San Diego, CA (2009).
- [3] H. H. Le, K.-A. Nguyen, Information-theoretic limits of health indicator construction for remaining useful life prediction, *Eng. Sci. Technol. Int. J.* 81 (2026) 102485. doi:10.1016/j.jestch.2026.102485.
- [4] F. Karimi Pour, D. Theilliol, V. Puig, G. Cembrano, Health-aware control design based on remaining useful life estimation for autonomous racing vehicle, *ISA Trans.* 113 (2021) 196–209. doi:10.1016/j.isatra.2020.03.032.
- [5] M. S. Félix, J. J. Martinez, C. Bérenguer, Remaining useful life (RUL) control of controlled systems under degradation, *Int. J. Robust Nonlinear Control* 35 (18) (2025) 8095–8116. doi:10.1002/rnc.70105.
- [6] J. C. Salazar, P. Weber, F. Nejari, R. Sarate, D. Theilliol, System reliability aware model predictive control framework, *Reliability Engineering & System Safety* 167 (2017) 663–672. doi:10.1016/j.res.2017.04.012.

- [7] P. Wen, S. Zhao, S. Chen, Y. Li, A generalized remaining useful life prediction method for complex systems based on composite health indicator, *Reliability Engineering & System Safety* 205 (2021) 107241. doi:10.1016/j.ress.2020.107241.
- [8] K. Liu, N. Z. Gebraeel, J. Shi, A data-level fusion model for developing composite health indices for degradation modeling and prognostic analysis, *IEEE Trans. Autom. Sci. Eng.* 10 (3) (2013) 652–664. doi:10.1109/TASE.2013.2250282.
- [9] C. Song, K. Liu, Statistical degradation modeling and prognostics of multiple sensor signals via data fusion: A composite health index approach, *IIE Trans.* 50 (10) (2018) 853–867. doi:10.1080/24725854.2018.1440673.
- [10] X. Fang, K. Paynabar, N. Gebraeel, Multistream sensor fusion-based prognostics model for systems with single failure modes, *Reliability Engineering & System Safety* 159 (2017) 322–331. doi:10.1016/j.ress.2016.11.008.
- [11] Y. Wang, I.-C. Lee, Y. Hong, X. Deng, Building degradation index with variable selection for multivariate sensory data, *Reliability Engineering & System Safety* 227 (2022) 108704. doi:10.1016/j.ress.2022.108704.
- [12] H. Li, Z. Zhang, X. Si, A dual-purpose data-model interactive framework for multi-sensor selection and prognosis, *Reliability Engineering & System Safety* 258 (2025) 110904. doi:10.1016/j.ress.2025.110904.
- [13] H. Yan, K. Liu, X. Zhang, J. Shi, Multiple sensor data fusion for degradation modeling and prognostics under multiple operational conditions, *IEEE Transactions on Reliability* 65 (3) (2016) 1416–1426. doi:10.1109/tr.2016.2575449.
- [14] L. Guo, N. Li, F. Jia, Y. Lei, J. Lin, A recurrent neural network based health indicator for remaining useful life prediction of bearings, *Neurocomputing* 240 (2017) 98–109. doi:10.1016/j.neucom.2017.02.045.
- [15] K. Javed, R. Gouriveau, N. Zerhouni, State of the art and taxonomy of prognostics approaches, trends of prognostics applications and open issues towards maturity at different technology readiness levels, *Mech. Syst. Signal Process.* 94 (2017) 214–236. doi:10.1016/j.ymssp.2017.01.050.
- [16] N. Li, N. Gebraeel, Y. Lei, X. Fang, X. Cai, T. Yan, Remaining useful life prediction based on a multi-sensor data fusion model, *Reliability Engineering & System Safety* 208 (2021) 107249. doi:10.1016/j.ress.2020.107249.
- [17] W. Q. Meeker, L. A. Escobar, *Statistical Methods for Reliability Data*, Wiley, 1998.
- [18] Z.-S. Ye, M. Xie, Stochastic modelling and analysis of degradation for highly reliable products, *Appl. Stoch. Models Bus. Ind.* 31 (1) (2015) 16–32. doi:10.1002/asmb.2063.
- [19] X.-S. Si, W. Wang, C.-H. Hu, D.-H. Zhou, Remaining useful life estimation — a review on the statistical data driven approaches, *Eur. J. Oper. Res.* 213 (1) (2011) 1–14. doi:10.1016/j.ejor.2010.11.018.
- [20] Z. Zhang, X. Si, C. Hu, Y. Lei, Degradation data analysis and remaining useful life estimation: a review on Wiener-process-based methods, *Eur. J. Oper. Res.* 271 (3) (2018) 775–796. doi:10.1016/j.ejor.2018.02.033.
- [21] L. Bian, N. Gebraeel, J. P. Kharoufeh, Degradation modeling for real-time estimation of residual lifetimes in dynamic environments, *IIE Transactions* 47 (5) (2015) 471–486. doi:10.1080/0740817x.2014.955153.
- [22] N. Li, N. Gebraeel, Y. Lei, L. Bian, X. Si, Remaining useful life prediction of machinery under time-varying operating conditions based on a two-factor state-space model, *Reliability Engineering & System Safety* 186 (2019) 88–100. doi:10.1016/j.ress.2019.02.017.
- [23] S. Jahani, S. Zhou, D. Veeramani, Stochastic prognostics under multiple time-varying environmental factors, *Reliability Engineering & System Safety* 215 (2021) 107877. doi:10.1016/j.ress.2021.107877.
- [24] Z. Lin, X. Liu, Y. Xiang, Y. Hong, Modeling multivariate degradation data with dynamic covariates under a bayesian framework, *Reliability Engineering & System Safety* 261 (2025) 111115. doi:10.1016/j.ress.2025.111115.
- [25] S.-S. Chen, X.-Y. Li, Modeling dependent degradation data considering inherent causal relationships for reliability analysis and remaining useful life prediction, *Reliability Engineering & System Safety* 277 (2026) 113080. doi:10.1016/j.ress.2026.113080.
- [26] Y. Langeron, A. Grall, A. Barros, A modeling framework for deteriorating control system and predictive maintenance of actuators, *Reliability Engineering & System Safety* 140 (2015) 22–36. doi:10.1016/j.ress.2015.03.028.
- [27] M. A. uit het Broek, R. H. Teunter, B. de Jonge, J. Veldman, Joint condition-based maintenance and condition-based production optimization, *Reliability Engineering & System Safety* 214 (2021) 107743. doi:10.1016/j.ress.2021.107743.
- [28] J. Zuo, C. Cadet, Z. Li, C. Bérenguer, R. Outbib, A deterioration-aware energy management strategy for the lifetime improvement of a multi-stack fuel cell

- system subject to a random dynamic load, *Reliability Engineering & System Safety* 241 (2024) 109660. doi:10.1016/j.ress.2023.109660.
- [29] A. Kamariotis, K. Tatsis, E. Chatzi, K. Goebel, D. Straub, A metric for assessing and optimizing data-driven prognostic algorithms for predictive maintenance, *Reliability Engineering & System Safety* 242 (2024) 109723. doi:10.1016/j.ress.2023.109723.
- [30] H. H. Le, K.-A. Nguyen, Generalization bounds and sample complexity for remaining useful life prediction from complete degradation trajectories, *Meas. Sci. Technol.* 37 (2026) 226203. doi:10.1088/1361-6501/ae7109.
- [31] H. Lim, B.-J. Yum, Optimal design of accelerated degradation tests based on Wiener process models, *J. Appl. Stat.* 38 (2) (2011) 309–325. doi:10.1080/02664760903406488.
- [32] K. A. Severson, P. M. Attia, N. Jin, N. Perkins, B. Jiang, Z. Yang, M. H. Chen, M. Aykol, P. K. Herring, D. Fragedakis, M. Z. Bazant, S. J. Harris, W. C. Chueh, R. D. Braatz, Data-driven prediction of battery cycle life before capacity degradation, *Nat. Energy* 4 (2019) 383–391. doi:10.1038/s41560-019-0356-8.
- [33] B. Wang, Y. Lei, N. Li, N. Li, A hybrid prognostics approach for estimating remaining useful life of rolling element bearings, *IEEE Trans. Rel.* 69 (1) (2020) 401–412. doi:10.1109/tr.2018.2882682.
- [34] P. Nectoux, R. Gouriveau, K. Medjaher, E. Ramasso, B. Chebel-Morello, N. Zerhouni, C. Varnier, PRONOSTIA: An experimental platform for bearings accelerated degradation tests, in: *IEEE Int. Conf. Prognostics Health Manage. (PHM)*, Denver, CO, USA, 2012, pp. 1–8.
- [35] B. Saha, K. Goebel, Battery data set, NASA Ames Prognostics Data Repository (2007).
- [36] A. Saxena, K. Goebel, D. Simon, N. Eklund, Damage propagation modeling for aircraft engine run-to-failure simulation, in: *Int. Conf. Prognostics Health Manage. (PHM)*, Denver, CO, USA, 2008, pp. 1–9. doi:10.1109/phm.2008.4711414.
- [37] M. Arias Chao, C. Kulkarni, K. Goebel, O. Fink, Aircraft engine run-to-failure dataset under real flight conditions for prognostics and diagnostics, *Data* 6 (1) (2021) 5. doi:10.3390/data6010005.
- [38] M. Arias Chao, C. Kulkarni, K. Goebel, O. Fink, Fusing physics-based and deep learning models for prognostics, *Reliability Engineering & System Safety* 217 (2022) 107961. doi:10.1016/j.ress.2021.107961.
- [39] T. M. Cover, J. A. Thomas, *Elements of Information Theory*, 2nd Edition, Wiley, 2006. doi:10.1002/047174882x.